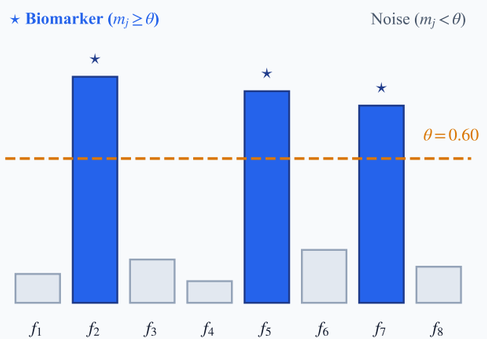
(a) Prior Profile & Seeding

Normalized relevance vector $\mathbf{m} \in [0,1]^D$



Continuous Coordinate Sampling

$$\mathbf{p}_i = \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1)$$

$\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ (Gaussian Perturbation)

$\mathbf{p}_1 = [0.14, 0.92, 0.16, 0.08, 0.85, 0.19, 0.84, 0.12]$

Step Mapping

$p_{i,j} \geq \theta$

(b) Deterministic Binarization

Binary decision matrix $\mathbf{X} \in \{0,1\}^{N \times D}$

	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	
$\mathbf{x}_1$	0	**1**	0	0	**1**	0	**1**	0	$d=3$
$\mathbf{x}_2$	0	**1**	0	0	**1**	0	0	0	$d=2$
$\mathbf{x}_3$	0	**1**	0	0	0	0	**1**	0	$d=2$
$\mathbf{x}_4$	0	0	0	0	**1**	0	**1**	0	$d=2$

Intrinsic Sparsity Advantage

Random Seeding: $d \approx D/2$ (Dense noise traps)

Prior Seeding: $d \ll D$ (Ultra-sparse core)

Early-stage evaluation cost suppressed by 96.8%

Evolution

$t_{\text{stag}} \geq \tau$

(c) Stagnation Diagnosis

Fitness sort upon stagnation ($t_{\text{stag}} \geq \tau$)

Superior Half (Top $N/2$ **Elite**)

$\mathbf{x}_1$: [0, 1, 0, 0, 1, 0, 1, 0] $f = 0.038$

Status: Global Best ($\mathbf{g}_{\text{best}}$) - Retained

$\mathbf{x}_2$: [0, 1, 0, 0, 1, 0, 0, 0] $f = 0.045$

Action: Preserve Intact (Protect Synergy)

50% Median Split

Inferior Half (Bottom $N/2$ **Stagnant**)

$\mathbf{x}_3$: [0, 1, 0, 0, 0, 0, 1, 0] $f = 0.089$

Status: Stagnant Trajectory Detected

$\mathbf{x}_4$: [0, 0, 0, 0, 1, 0, 1, 0] $f = 0.098$

Action: Flagged for Adaptive Refresh

Recovery

$t_{\text{stag}} \leftarrow 0$

(d) Adaptive Re-injection

Exploratory diversity restored ($t_{\text{stag}} \leftarrow 0$)

Retained Elite Sub-Swarm

$\mathbf{x}_1$: [0, 1, 0, 0, 1, 0, 1, 0] **Locked**

Preserved historical search momentum

$\mathbf{x}_2$: [0, 1, 0, 0, 1, 0, 0, 0] **Locked**

Verified biomarker core protected

Resumed Dual Co-evolution

Refreshed Prior Sub-Swarm

$\mathbf{x}_3^{\text{new}}$**: [0, 1, 0, 1, 1, 0, 0, 0]** **Resampled**

Perturbation: $\mathbf{m} + \boldsymbol{\varepsilon}_3 \rightarrow$ New synergy (f_4 active)

$\mathbf{x}_4^{\text{new}}$**: [0, 1, 0, 0, 1, 0, 1, 1]** **Resampled**

Perturbation: $\mathbf{m} + \boldsymbol{\varepsilon}_4 \rightarrow$ New synergy (f_8 active)